Connor Nieh

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Design Portfolio: https://connornieh.github.io/med-device-portfolio/

EDUCATION

Duke University, M.Eng in Medical Technology Design *Expected Dec. 2026*
- Coursework: Advanced Design and Manufacturing, Design Health, Signal Processing, Machine Learning and Imaging, Quality Management Systems

University of California, Riverside, B.S. Bioengineering Sept. 2021 – June 2025
- Coursework: Bioinstrumentation, Biomaterials, Biomechanics, Circulation Physiology, C++, Electrical Engineering, Optical Imaging, Statistics, Tissue Engineering, Quantitative Physiology
- Awards: Cum Laude, Dean's Honors List, Chancellor's Honors List, 3.76 GPA

MEDICAL DEVICE DEVELOPMENT EXPERIENCE

Design Health Project, Duke Design in Healthcare Series Aug. 2025 – Present
- Engaging in ethnographic research and clinical shadowing to define unmet needs in cardiothoracic operating rooms
- Translating user needs into engineering requirements and developing CAD design concepts for rapid prototyping
- Performing bench-top testing to quantitatively evaluate design features and validate product requirements
- Applying root cause analysis and the MECE framework to systematically evaluate and filter engineering concepts

Graduate Research Assistant, BIOS Lab – Duke University Sept. 2025 – Present
- Leading R&D efforts to translate quantitative phase imaging into a low-cost, diagnostic device for sickle cell disease
- Defining system-level design requirements and performance metrics to support clinical validation
- Implementing machine learning workflows to process and analyze high-throughput flow cytometry experiments
- Iteratively developed and optimized a repeatable manufacturing protocol, increasing microfluidic chip yield by 500%

Microscope Integrated Instrumentation for OCT Project, ONI Lab - UCR Sept. 2024 – June 2025
- Designed, prototyped, and verified an optomechanical interface to enable multimodal OCT imaging in a microscope
- Generated CAD models and technical drawings to support fabrication and assembly of optical housing components
- Conducted verification testing to optimize imaging resolution, signal alignment, and optical collimation
- Improved device reliability through precise optomechanical alignment and tolerance analysis

Medical Device Development Intern, Keck Graduate Institute June 2023 – July 2023
- Co-designed and prototyped a medical device to promote neuroplasticity in dementia patients, integrating mechanical, electrical, and software subsystems to deliver a functional, patient-focused product
- Translated user and clinical needs into functional specifications and early-stage design inputs
- Led an interdisciplinary team through regulatory, quality management system, and R&D pathways

RESEARCH EXPERIENCE

Undergraduate Researcher, TIME Lab - UCR Oct. 2023 – June 2025
- Investigated stress tolerance of genetically engineered stem cell therapies for regenerative medicine
- Developed a computational multi-cell proliferation model in MATLAB to simulate cell-cycle dynamics
- Presented findings on ASC toxic responses at the UCR Undergraduate Research Symposium

Summer Biomedical Intern, Terasaki Institute for Biomedical Innovation June 2024 – Sept. 2024
- Led parallel projects in iPSC characterization, 3-D cell modeling, and drug assay development, accelerating experiment throughput and supporting precision medicine studies in oncology
- Developed and documented 5 tissue processing, organoid culturing, cell harvesting, and drug dosing protocols
- Presented research on novel organoid-based disease modeling and diagnostic platform applications

SKILLS

R&D: CAD Modeling, Design Documentation, DFM, DHF Documentation, Ethnographic Research, FEA, Iterative Prototyping, MECE Analysis, Requirements Specification, Root Cause Analysis, Validation Testing, QMS, 3-D Printing
Software: Arduino Programming, COMSOL Multiphysics, Figma, Fusion 360, Google Suite, Image Reconstruction, MATLAB, Microsoft Suite, On-Shape CAD, Python, PyTorch, SOLIDWORKS, 3-D Slicer
Laboratory: Clean Room Protocols, Confocal Microscopy, DNA/RNA Isolation, ELISA, Fluorescence Imaging, Microfluidic Fabrication, OCT Imaging, Optics, PCR, Quantitative Phase Imaging, 2-D/3-D Cell Culture

EXTRACURRICULARS

International Society of Pharmaceutical Engineering Mar. 2023 – June 2025
Vice President, University of California, Riverside
- Coordinated member participation in professional workshops and networking events with industry leaders